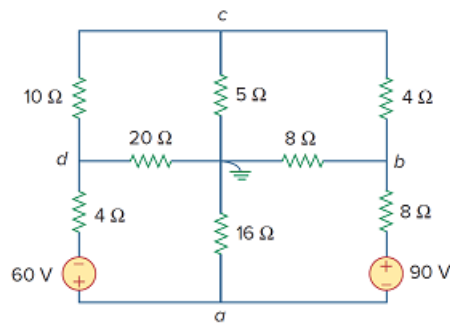


Problem

Use *MATLAB* to find the voltages at nodes *a*, *b*, *c*, and *d* in the circuit of Fig. 3.77.

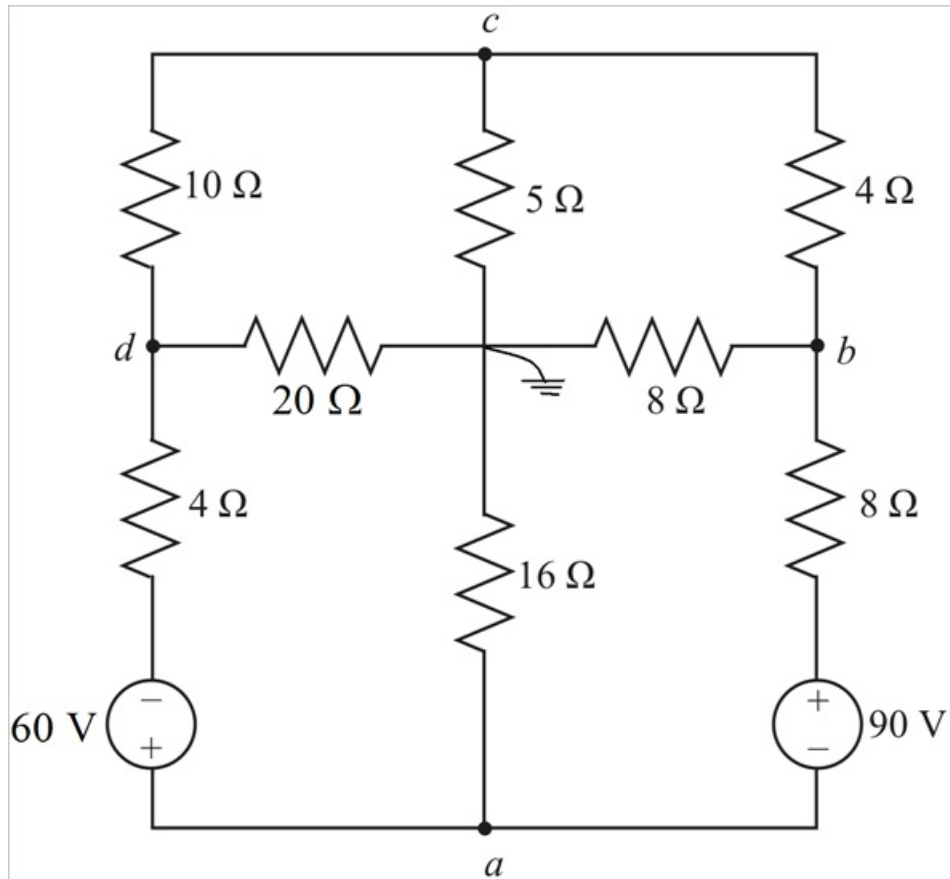
Figure 3.77:



Step-by-step solution

Step 1 of 6

The given circuit is



Step 2 of 6

By applying the KCL at node c we get

$$\frac{v_d - v_c}{10} = \frac{v_c - v_b}{4} + \frac{v_c}{5}$$

$$\frac{v_c - v_b}{4} + \frac{v_c}{5} - \frac{v_d - v_c}{10} = 0$$

$$\frac{5v_c - 5v_b + 4v_c - 2v_d + 2v_c}{20} = 0$$

$$11v_c - 5v_b - 2v_d = 0$$

$$-5v_b + 11v_c - 2v_d = 0 \dots\dots (1)$$

Step 3 of 6

By applying the KCL at node b we get

$$\frac{v_a + 90 - v_b}{8} + \frac{v_c - v_b}{4} = \frac{v_b}{8}$$

$$\frac{v_a + 90 - v_b}{8} + \frac{v_c - v_b}{4} - \frac{v_b}{8} = 0$$

$$\frac{v_a + 90 - v_b + 2v_c - 2v_b - v_b}{8} = 0$$

$$v_a - 4v_b + 2v_c = -90 \dots\dots (2)$$

Step 4 of 6

By applying the KCL at node a we get

$$\frac{v_a - 60 - v_d}{4} + \frac{v_a}{16} + \frac{v_a + 90 - v_b}{8} = 0$$

$$\frac{4v_a - 240 - 4v_d + v_a + 2v_a + 180 - 2v_b}{16} = 0$$

$$7v_a - 2v_b - 4v_d = 60 \dots\dots\dots (3)$$

Step 5 of 6

By applying the KCL at node d we get

$$\frac{v_a - 60 - v_d}{4} = \frac{v_d}{20} + \frac{v_d - v_c}{10}$$

$$\frac{v_a - 60 - v_d}{4} - \frac{v_d}{20} - \frac{v_d - v_c}{10} = 0$$

$$\frac{10v_a - 600 - 10v_d - 2v_d - 4v_d + 4v_c}{40} = 0$$

$$10v_a + 4v_c - 16v_d - 600 = 0$$

$$5v_a + 2v_c - 8v_d = 300 \dots\dots\dots(4)$$

Step 6 of 6

Thus we have 4 equations from 4 nodes

$$-5v_b + 11v_c - 2v_d = 0$$

$$v_a - 4v_b + 2v_c = -90$$

$$7v_a - 2v_b - 4v_d = 60$$

$$5v_a + 2v_c - 8v_d = 300$$

The above equations can be defined in matrix form as

$$\begin{bmatrix} 0 & -5 & 11 & -2 \\ 1 & -4 & 2 & 0 \\ 7 & -2 & 0 & -4 \\ 5 & 0 & 2 & -8 \end{bmatrix} \begin{bmatrix} v_a \\ v_b \\ v_c \\ v_d \end{bmatrix} = \begin{bmatrix} 0 \\ -90 \\ 60 \\ 300 \end{bmatrix}$$

Using Matlab:-

```
>> a= [0 -5 11 -2; 1 -4 2 0; 7 -2 0 -4; 5 0 2 -8];
```

```
>> b= [0;-90; 60; 300];
```

```
>> c=inv(a)*b
```

c =

-10.5556

20.5556

1.3889

-43.7500